

Skills Summary

Exponential Form and Logarithmic Form

Use this reference while completing your worksheet or when studying.

Skill 1 — Exponential statement to logarithm

The one fact, two ways: $b^y = x \iff \log_b x = y$ (with $b > 0$, $b \neq 1$, $x > 0$).
The base b never moves. The *exponent* becomes the value of the logarithm, and the *result* becomes its input.

Example: Rewrite $3^4 = 81$ in logarithmic form.

Step 1. The statement already names a base, an exponent and a result, so the switch is just a matter of putting each in its new place.

Step 2. Base 3 stays the base; the exponent 4 becomes the value; the result 81 becomes the input.

$$\Rightarrow \log_3 81 = 4$$

Try it: Rewrite $2^6 = 64$ in logarithmic form.

check: $\log_2 64 = 6$

Skill 2 — Logarithm back to a power

Method: read $\log_b x = y$ out loud as “ b to the power y gives x ”, then write $b^y = x$ and evaluate. This is how you release an unknown that is trapped inside a logarithm.

Example: Solve $\log_4 x = 3$.

Step 1. The unknown is the *input* of the logarithm, so switching to exponential form puts it alone on one side.

Step 2. $\log_4 x = 3$ becomes $4^3 = x$, and $4^3 = 64$.

$$\Rightarrow x = 64$$

Try it: Solve $\log_3 x = 4$.

check: $\log_3 81 = 4$

Skill 3 — Evaluating a logarithm

Ask the question the notation asks: $\log_b x$ means “what power of b gives x ?” Build up the powers of b until you reach x . The answer can be a fraction (a root) or negative (a reciprocal).

Example: Evaluate $\log_5 125$.

Step 1. Evaluating a logarithm is a search through the powers of its base, so list them until the input appears.

Step 2. $5^1 = 5$, $5^2 = 25$, $5^3 = 125$, so the exponent that works is 3.

$$\Rightarrow 3$$

Try it: Evaluate $\log_8 2$.

check: $\frac{8}{1}$

Skill 4 — Solving by switching form

Method: when the unknown is inside a logarithm, switch to exponential form first, then solve the ordinary equation left behind. Check afterwards that the logarithm's input really is positive at your answer.

Example: Solve $\log_3(x + 2) = 2$.

Step 1. The unknown sits inside the logarithm, so nothing algebraic can reach it until the statement is written as a power.

Step 2. $3^2 = x + 2$, so $9 = x + 2$. The input at $x = 7$ is $9 > 0$, so the answer stands.

$$\Rightarrow x = 7$$

Try it: Solve $\log_5(2x - 1) = 2$.

check: $13 = x$